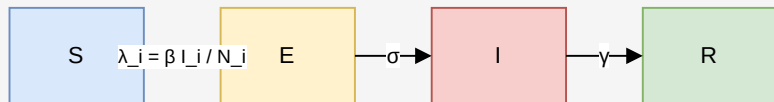


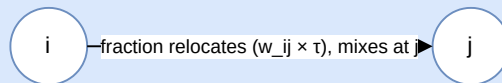
Metapopulation reaction–diffusion, per node per day

City i — local REACTION (SEIR shown; SIR/SIS/SQIR are variants)

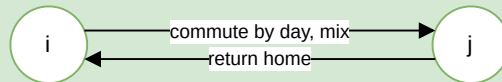


SIS: $I \rightarrow S(\gamma)$ · SQIR: $I \rightarrow Q(\kappa) \rightarrow R(\gamma_Q)$

DIFFUSIVE mobility (air, water)



RECURRENT mobility (land / commuting)



couples force of infection:
 $\pi_i = \sum_j C_{ij} \cdot I^*_j / N^*_j$

DAILY LOOP: reaction (all nodes) → mobility → next day

No land $\Rightarrow C = I \Rightarrow \pi_i = I_i / N_i$ (recovers plain diffusive). Recurrent vs diffusive commuting $\approx 5\times$ lower peak on Europe air+land.